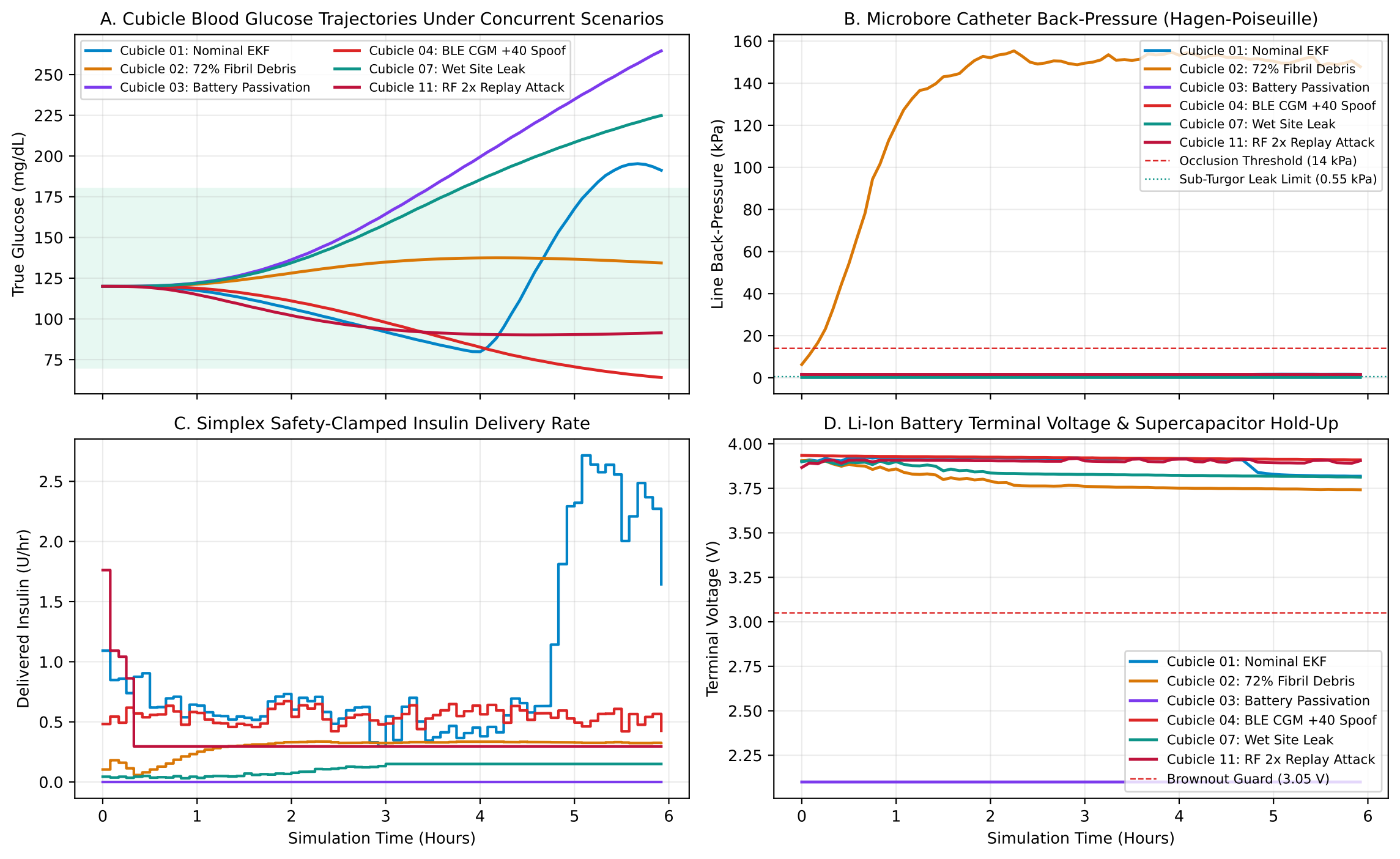
A. Cubicle Blood Glucose Trajectories Under Concurrent Scenarios
True Glucose (mg/dL)
Cubicle 01: Nominal EKF
Cubicle 02: 72% Fibril Debris
Cubicle 03: Battery Passivation
Cubicle 04: BLE CGM +40 Spoof
Cubicle 07: Wet Site Leak
Cubicle 11: RF 2x Replay Attack
B. Microbore Catheter Back-Pressure (Hagen-Poiseuille)
Line Back-Pressure (kPa)
Occlusion Threshold (14 kPa)
Sub-Turgor Leak Limit (0.55 kPa)
C. Simplex Safety-Clamped Insulin Delivery Rate
Delivered Insulin (U/hr)
Simulation Time (Hours)
D. Li-Ion Battery Terminal Voltage & Supercapacitor Hold-Up
Terminal Voltage (V)
Brownout Guard (3.05 V)
Simulation Time (Hours)